SECTION A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			$Mg^{2+}(g) + 2Cl^{-}(g) \rightarrow MgCl_2(s)$		1		1		
	(b)			standard enthalpy change of hydration	1			1		
2	(a)			oxidation state of chlorine at start is 0; at end it is –1 (in NaCl) and +1 (in NaOCl) (1) chlorine (or same element) has been both oxidised and reduced (1)	1	1		2		
	(b)			bleach	1			1		
3				$[H^+] = \frac{1.00 \times 10^{-14}}{2 \times 2.34 \times 10^{-2}} = 2.13 \times 10^{-13}$ mol dm^{-3} (1) pH = – log 2.13×10^{-13} = 12.7 (1) ecf possible		2		2	2	
4				more efficient release of energy from fuel	1			1		
5				+3 (must give reason to gain this mark) (1) inert pair effect increases down the group (so oxidation state of [group number – 2] is more stable) (1)	1	1		2		
				Section A total	**5**	**5**	**0**	**10**	**2**	**0**